

$$1.) M_m = 1500 \text{ Nm}$$

$$S_m = \frac{My}{I} = \frac{1500 \times r}{\frac{\pi r^4}{4}}$$

$$= \frac{6000}{\pi r^3}$$

$$S_a = \frac{My}{I} = \frac{1000 \times r}{\frac{\pi r^4}{4}}$$

$$= \frac{4000}{\pi r^3}$$

Using the Soderberg criteria:

$$\frac{S_a(SF)}{S_e} + \frac{S_m}{S_y} \leq 1$$

$$\frac{2.5 \left(\frac{4000}{\pi r^3} \right)}{170 \times 10^6} + \frac{\frac{6000}{\pi r^3}}{210 \times 10^6} \leq 1$$

$$r^3 \geq \frac{2.5(4000)}{170 \times 10^6 \pi} + \frac{6000}{210 \times 10^6 \pi}$$

$$\frac{d^3}{8} \geq 2.781867913 \times 10^{-5}$$

$$d \geq 0.0606004092 \text{ m}$$

$$\geq 60.6 \text{ mm}$$

iii) If the fatigue strength reduction factor, $K_f = 2$, then the Soderberg relation becomes

$$\frac{S_a(SF)}{\frac{S_e}{K_f}} + \frac{S_m}{S_y} \leq 1$$

$$\frac{d^3}{8} \geq \frac{2.5(4000)}{\frac{1}{2}(170 \times 10^6 \text{ N})} + \frac{6000}{210 \times 10^6 \text{ N}}$$

$$d^3 \geq 3.723423206 \times 10^{-4}$$

$$d \geq 71.94171728 \times 10^{-3}$$

$$\geq 71.94 \text{ mm}$$

$$2) N_f = \frac{2}{C(YS_R)^m \pi^{\frac{m}{2}} (2-m)} \left(a_f^{1-\frac{m}{2}} - a_o^{1-\frac{m}{2}} \right)$$

$$N_f = 10^6$$

$$Y = 1$$

$$K_c = 1.8 \times 10^6$$

$$m = 2.513$$

$$C = 2 \times 10^{-6}$$

$$a_o = \frac{40 \times 10^{-6}}{2} = 20 \times 10^{-6}$$

$$K_c = Y \sigma \sqrt{\pi a_f}$$

$$da_f = \frac{1}{\pi} \frac{K_c^2}{Y^2 \sigma^2}$$

Due to the fully reversed cycle, only the tensile part of the cycle is responsible for crack growth

$$\therefore S_R = \sigma$$

2)

$$N_f = \frac{2}{C(YSR)^m \pi^{\frac{m}{2}} (2-m)} \left(a_f^{1-\frac{m}{2}} - a_o^{1-\frac{m}{2}} \right)$$

$$10^6 = \frac{2}{2 \times 10^{-6} (1\sigma)^{2.513} \pi^{\frac{2.513}{2}} (2-2.513)} \left[\left(\frac{1}{\pi} \left(\frac{1.8}{1\sigma} \right)^2 \right)^{1-\frac{2.513}{2}} - (20 \times 10^{-6})^{1-\frac{2.513}{2}} \right]$$

$$10^6 = - \frac{2}{4.323307993 \times 10^{-6} \sigma^{2.513}} \left[\frac{0.9921198689}{\sigma - 0.513} - 16.0430082 \right]$$

$$-2.161653997 \sigma^{2.513} = 0.9921198689 \sigma^{-0.513} - 16.04300822$$

Solving the equation:

$$\sigma = 2.13723348 \text{ MPa}$$

$$\approx 2.14 \text{ MPa}$$

$$3) S_R = 500 \text{ MPa}$$

$$\sigma = S_{\max} = 500 \text{ MPa}$$

$$N_f = 365 \times 25$$

$$= 9125$$

$$K_c = Y \sigma \sqrt{\pi a_f}$$

$$a_f = \frac{1}{\pi} \left(\frac{K_c}{Y \sigma} \right)^2$$

i) For maraging steel:

$$N_f = \frac{2}{C(Y S_R)^m \pi^{\frac{m}{2}} (2-m)} \left(a_f^{1-\frac{m}{2}} - a_0^{1-\frac{m}{2}} \right)$$

$$9125 = \frac{2}{0.15 \times 10^{-11} (1.12 \times 500)^{4.1} \pi^{\frac{4.1}{2}} (2-4.1)} \left[\left(\frac{1}{\pi} \left(\frac{82}{1.12(500)} \right)^2 \right)^{1-\frac{4.1}{2}} - a_0^{1-\frac{4.1}{2}} \right]$$

$$-27812.59943 = 188.0152745 - a_0^{-1.05}$$

$$a_0 = 58.15704324 \times 10^{-6} \text{ m}$$

$$\approx 58.2 \mu\text{m}$$

3:ii) For medium strength steel:

$$N_f = \frac{2}{C(YS_R)^m \pi^{\frac{m}{2}} (2-m)} \left(a_f^{1-\frac{m}{2}} - a_0^{1-\frac{m}{2}} \right)$$

$$9125 = \frac{2}{0.24 \times 10^{-11} (1.12 \times 500)^{3.3} \pi^{\frac{3.3}{2}} (2-3.3)} \left[\left(\frac{1}{\pi} \left(\frac{50}{1.12(500)} \right)^2 \right)^{1-\frac{3.3}{2}} - a_0^{1-\frac{3.3}{2}} \right]$$

$$-110.3253339 = 48.65537753 - a_0^{-0.65}$$

$$a_0 = 410.5029213 \times 10^{-6} \text{ m}$$

$$\approx 411 \mu\text{m}$$

Since the initial crack length permissible, a_0 , for medium strength steel (411 μm) is more than that of maraging steel, we conclude that based on fatigue considerations, the medium strength steel is more tolerant to initial defects.

Since fatigue is caused by cyclic loading, we can conclude that under cyclic loading conditions, the medium strength steel is the better material.

3) Maraging steel:

$$a_f = \frac{1}{\pi} \left(\frac{K_c}{Y_\sigma} \right)^2$$

$$\begin{aligned} a_f &= \frac{1}{\pi} \left(\frac{82}{1.12 \times 500} \right)^2 \\ &= 6.824986208 \times 10^{-3} \text{ m} \\ &\approx 6.82 \text{ mm} \end{aligned}$$

Medium strength steel:

$$a_f = \frac{1}{\pi} \left(\frac{K_c}{Y_\sigma} \right)^2$$

$$\begin{aligned} &= \frac{1}{\pi} \left(\frac{56}{1.12 \times 500} \right)^2 \\ &= 2.537546924 \times 10^{-3} \\ &\approx 2.54 \text{ mm} \end{aligned}$$

Based purely on fracture mechanics, our calculation gave the permissible crack length, a_f , for the maraging steel as 6.82 mm, which is more than that of the medium strength steel (2.54 mm).

Thus, based on fracture mechanics considerations, the maraging steel is more damage-resistant. In other words, we can say that under stated loading conditions, the maraging steel is the better material.

$$4) \frac{6 \times 10^5}{2 \times 10^6} + \frac{150 \times 10^3}{5 \times 10^5} + \frac{n}{125 \times 10^3} = 1$$

$$n = 50,000 \text{ cycles}$$

5)

Number of cycles (n)	Stress	N	$\frac{n}{N}$
105	50	4.22×10^{15}	$2.371373706 \times 10^{-11}$
104	100	8.34×10^{10}	$1.19837759 \times 10^{-7}$
500	150	147904162	$3.38056748 \times 10^{-6}$
200	180	8565908.392	$2.334837017 \times 10^{-5}$
100	185	5582748.473	$1.79123241 \times 10^{-5}$
50	200	1651249.258	$3.028010444 \times 10^{-5}$
10	210	770424.0607	$1.297986461 \times 10^{-5}$

$$\sum \left(\frac{n}{N} \right) = 8.802109228 \times 10^{-5}$$

$$\text{Number of flights} = \frac{1}{8.802109228 \times 10^{-5}}$$

$$= 11360.91332$$

$$\approx 11360$$